

COMP6237 Data Mining

Lecture 9: Market Basket Analysis

Zhiwu Huang

Zhiwu.Huang@soton.ac.uk

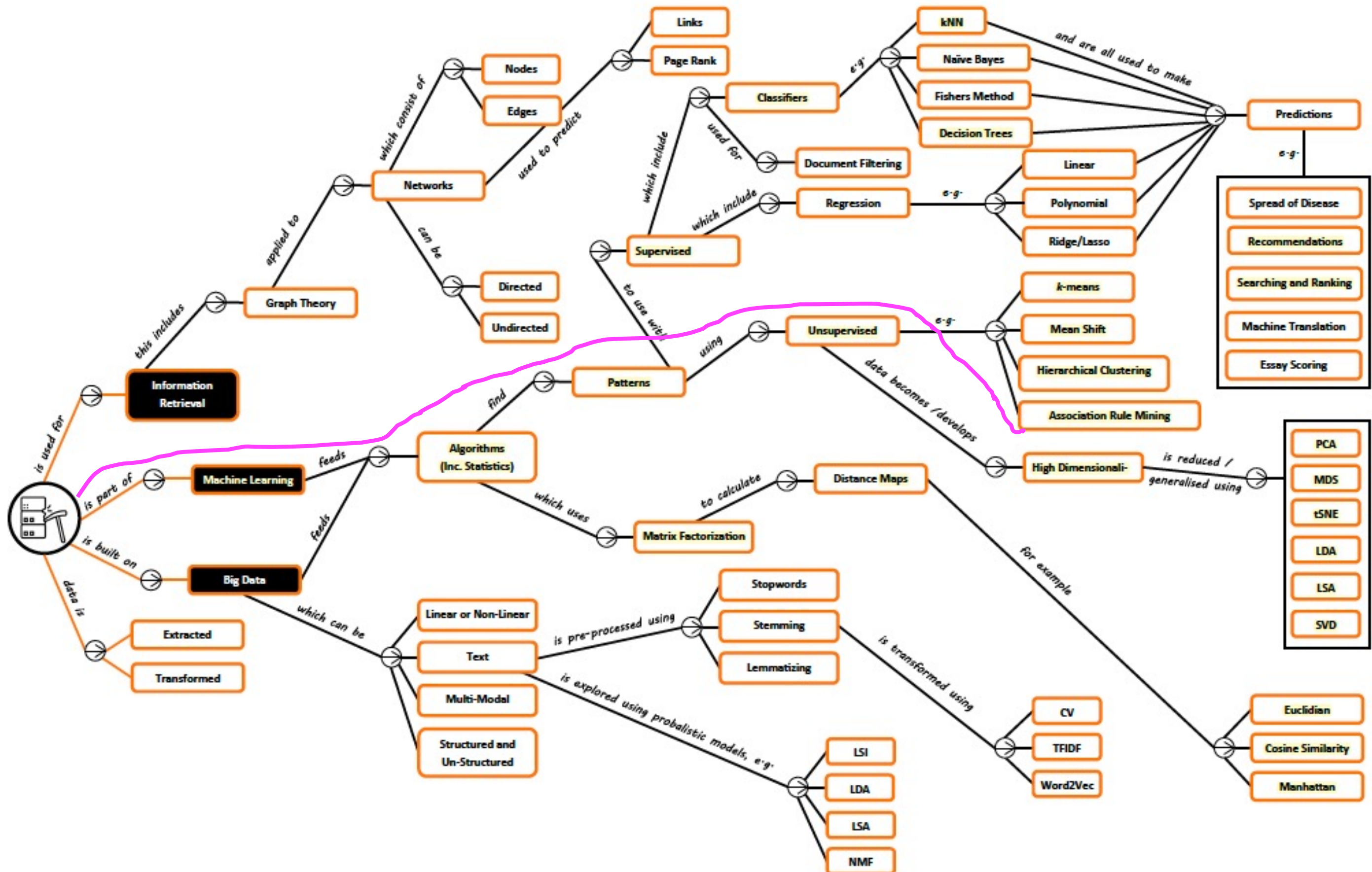
Lecturer (Assistant Professor) @ VLC of ECS
University of Southampton

Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Market Basket – Roadmap



Market Basket – Textbook

5 Association Analysis: Basic Concepts and Algorithms

*Many business enterprises accumulate large quantities of data from their day-to-day operations. For example, huge amounts of customer purchase data are collected daily at the checkout counters of grocery stores. **Table 5.1**  gives an example of such data, commonly known as **market basket transactions**. Each row in this table corresponds to a transaction, which contains a unique identifier labeled TID and a set of items bought by a given customer. Retailers are interested in analyzing the data to learn about the purchasing behavior of their customers. Such valuable information can be used to support a variety of business-related applications such as marketing promotions, inventory management, and customer relationship management.*

- ▶ Introduction to Data Mining, *P. Tan et al*
<https://www-users.cse.umn.edu/~kumar001/dmbook/index.php>

Market Basket – Overview (1/4)

Why analyse market baskets?

Get insight:

- ▶ do products sell quickly or slowly
- ▶ which products are sold together?
- ▶ which might need a promotion?

Use that to take action:

- ▶ store layout
- ▶ promotions
- ▶ recommendations

Market Basket – Overview (2/4)



Beer and Nappies

Back in 1992 A data consultant was using SQL queries to find things were often bought along side nappies (Diapers in the US), as nappies are high margin, they wanted to sell more of them.

They were looking to find things to put on the shelves near each other. She found a correlation between beer sales, and nappy sales, and emailed her colleagues about it.

There was no good statistical basis for this link, but the story has become well known, one of the first to 'go viral'

Market Basket – Overview (3/4)

Market Basket analysis:

Given a database of transactions

Find groups of items that are frequently bought together



Each transaction is a set of items, a basket, called here an *itemset*

This allows companies to understand why people make certain purchases

Market Basket – Overview (4/4)

EXAMPLE OF ASSOCIATION RULES



Assume there are 5 customers

3 of them bought **milk**, 2 bought **potato chip** and 2 bought **both of them**

Transaction 1: Frozen pizza, cola, milk
Transaction 2: Milk, potato chips
Transaction 3: Cola, frozen pizza
Transaction 4: Milk, potato chips
Transaction 5: Cola, pretzels

milk → **potato chip**

✓ support milk = $P(\text{milk}) = 3/5 = 0.6$

support potato chip = $P(\text{potato chip}) = 2/5 = 0.4$

support = $P(\text{milk \& potato chip}) = 2/5 = 0.4$

✓ **confidence**

= support (milk & potato chip) / support(milk)

= $0.4/0.6$

= 0.67

CONFIDENCE = $P(\text{Milk \& potato chip}) / P(\text{Milk})$



How about
Potato chip
→ **Milk** ?

Market Basket – Learning Outcomes

- **LO1:** Demonstrate an understanding of market basket analysis concepts and techniques, such as: (exam)
 - ❖ Calculating support and confidence for itemsets
 - ❖ Understanding the key steps of the Apriori algorithm for association rule mining
 - ❖ Using the Apriori algorithm to generate association rules from transaction data
- **LO2:** Implement the learned algorithms using associate rule mining algorithms (coursework)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*

Market Basket – Definition

- Market basket transaction data:
 - t_1 : {bread, cheese, milk}
 - t_2 : {apple, eggs, salt, yogurt}
 - ...
 - t_n : {biscuit, eggs, milk}
- Concepts:
 - An *item*: an item/article in a basket (i)
 - I : the set of *all items* sold in the store ($\{i_1, i_2, \dots, i_m\}$)
 - A *transaction*: items purchased in a basket; it may have TID (transaction ID) (t)
 - A *transactional dataset*: A set of transactions ($T = \{t_1, t_2, \dots, t_n\}$)

Market Basket – Definition

- An association rule is an implication of the form:

$$X \rightarrow Y, \text{ where } X, Y \subset I, \text{ and } X \cap Y = \emptyset$$

- $I = \{i_1, i_2, \dots, i_m\}$: a set of *all items*
- An itemset X is a set of items, where $X \subset I$.
 - E.g., $X = \{\text{milk, bread}\}$ is an itemset.
 - A k -itemset is an itemset with k items.
 - E.g., $\{\text{milk, bread, cereal}\}$ is a 3-itemset
- A transaction t contains an itemset X , if $X \subseteq t, X \subset I$

Market Basket – Definition

- **Support count:** The support count of an itemset A , denoted by $A.count$, in a data set T is the number of transactions in T that contain A . Assume T has n transactions.
- Then, support and confidence for the rule $X \rightarrow Y$

$$support = \frac{(X \cup Y).count}{n}$$

number of
transactions that
simultaneously
contains X and Y
total number of
transactions

$$confidence = \frac{(X \cup Y).count}{X.count}$$

number of transactions
that only contain
itemset X

Market Basket – Definition

- Minimum Support Threshold: $min_sup = s\%$ (e.g., 40%)
- Minimum Confidence threshold: $min_conf = c\%$ (e.g., 60%)

Market Basket – Definition

- **Frequent itemset**

- Suppose *min_sup* is the minimum support threshold
- An itemset satisfies minimum support if the occurrence frequency of the itemset is greater or equal to *min_sup*
- If an itemset satisfies minimum support, then it is a frequent itemset

Itemset: A set of items is referred to as itemset; An itemset containing k items is called **k-itemset**

Market Basket – Definition

Rules that satisfy both a *minimum support* threshold and a *minimum confidence* threshold are called **strong rules**

Market Basket – Example

- **Itemset**
- **Support count (σ)**
- **Support (s)**
- **Frequent Itemset**

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Market Basket – Example

- **Itemset**
 - A collection of one or more items
 - Example: {Milk, Bread, Diaper}
 - *k*-itemset
 - An itemset that contains *k* items
 - 3-itemset: {Milk, Diaper, Beer}
- **Support count (σ)**
- **Support (*s*)**
- **Frequent Itemset**

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
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Market Basket – Example

- **Itemset**

- A collection of one or more items
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5	Bread, Milk, Diaper, Coke

- **Support count (σ)**

- Frequency of occurrence of an itemset
- E.g. $\sigma(\{\text{Milk, Diaper, Beer}\}) = 2$

- **Support (s)**

- **Frequent Itemset**

Market Basket – Example

- **Itemset**

- A collection of one or more items
 - Example: {Milk, Bread, Diaper}
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- **Support count (σ)**

- Frequency of occurrence of an itemset
- E.g. $\sigma(\{\text{Milk, Diaper, Beer}\}) = 2$

- **Support (s)**

- Fraction of transactions that contain an itemset
- E.g. $s(\{\text{Milk, Diaper, Beer}\}) = 2/5$

- **Frequent Itemset**

Market Basket – Example

- **Itemset**

- A collection of one or more items
 - Example: {Milk, Bread, Diaper}
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- **Frequent Itemset**

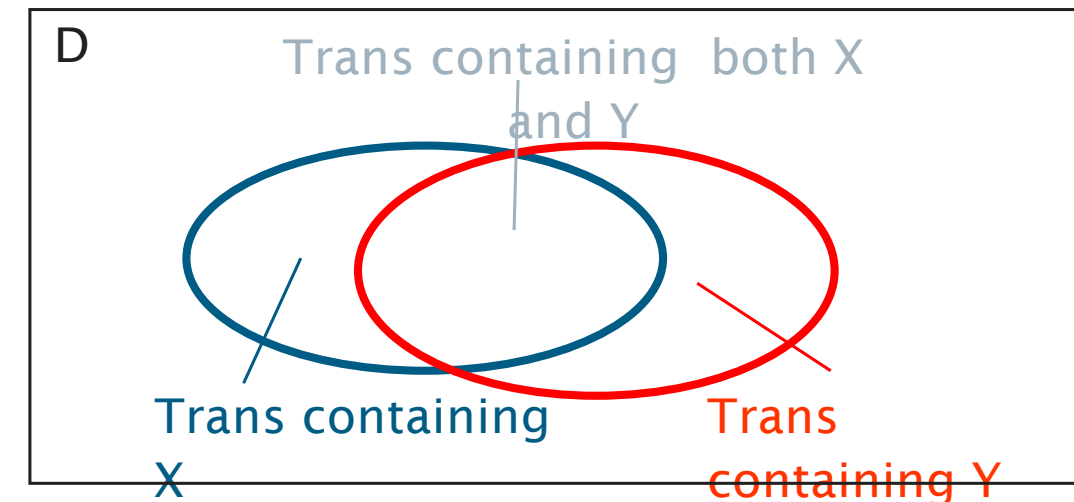
- An itemset whose support is greater than or equal to a *minsup* threshold

- An association rule r is **strong** if
 - $Support(r) \geq min_sup$
 - $Confidence(r) \geq min_conf$
- Given a set of transactions T , the goal of association rule mining is to find all rules having
 - support $\geq min_sup$ threshold
 - confidence $\geq min_conf$ threshold

Market Basket – Example

- **Association Rule**
 - An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets, and $X \cap Y = \emptyset$
 - Example: $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

Antecedent $\rightarrow$ Consequent



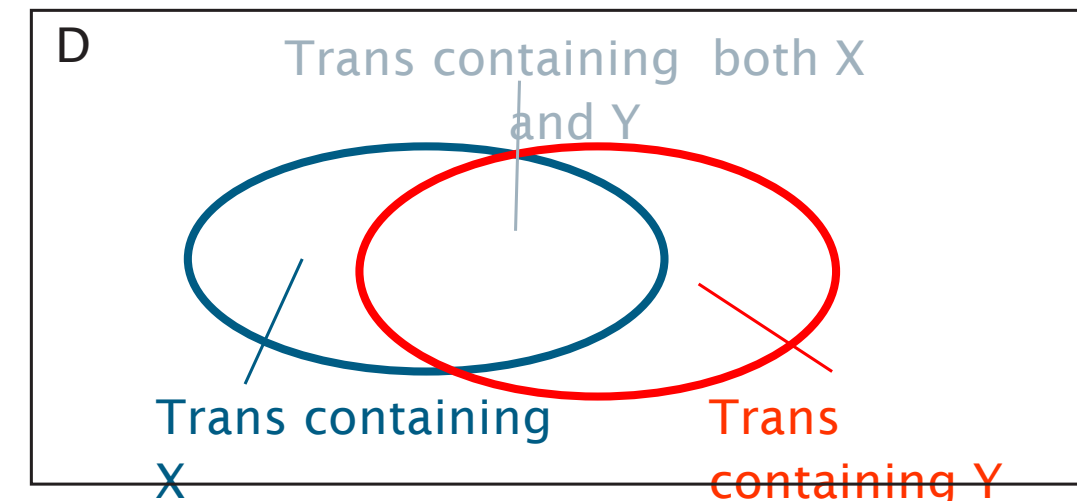
TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Rule: $\{\text{Milk, Diaper}\} \Rightarrow \text{Beer}$

Market Basket – Example

- **Association Rule**
 - An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets, and $X \cap Y = \emptyset$
 - Example: $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$
- **Rule Evaluation Metrics**
 - **Support (s)**
 - ◆ Fraction of transactions that contain both X and Y

$$P(X \cup Y) = \frac{\#transcontaining(X \cup Y)}{\#transinD}$$



TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Rule: $\{\text{Milk, Diaper}\} \Rightarrow \text{Beer}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

Market Basket – Example

- Association Rule

- An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets, and $X \cap Y = \emptyset$
- Example: {Milk, Diaper} $\rightarrow$ {Beer}

- Rule Evaluation Metrics

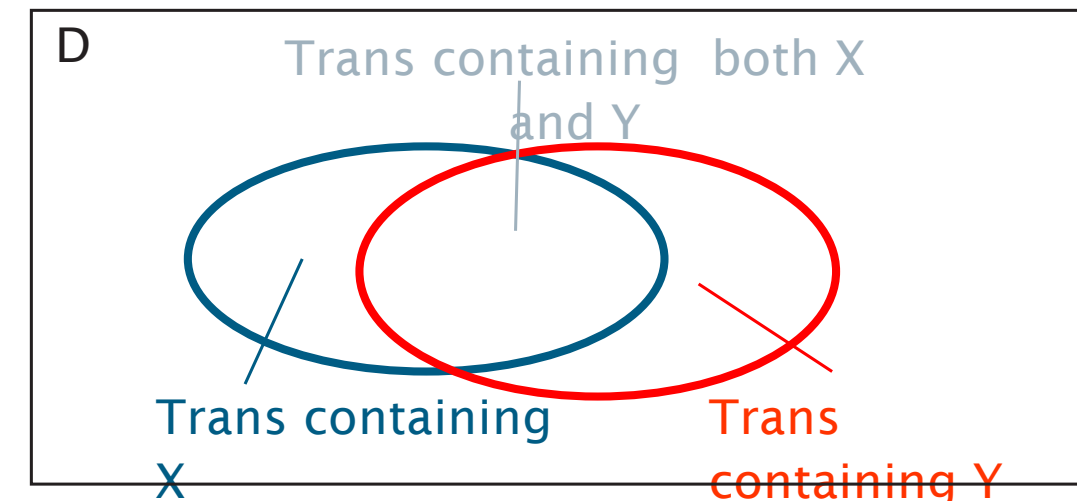
- Support (s)
 - ◆ Fraction of transactions that contain both X and Y

$$P(X \cup Y) = \frac{\#transcontaining(X \cup Y)}{\#transinD}$$

- Confidence (c)

- ◆ Measures how often items in Y appear in transactions that contain X

$$P(Y|X) = \frac{\#transcontaining(X \cup Y)}{\#transcontainingX}$$



TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Rule: {Milk, Diaper} $\Rightarrow$ Beer

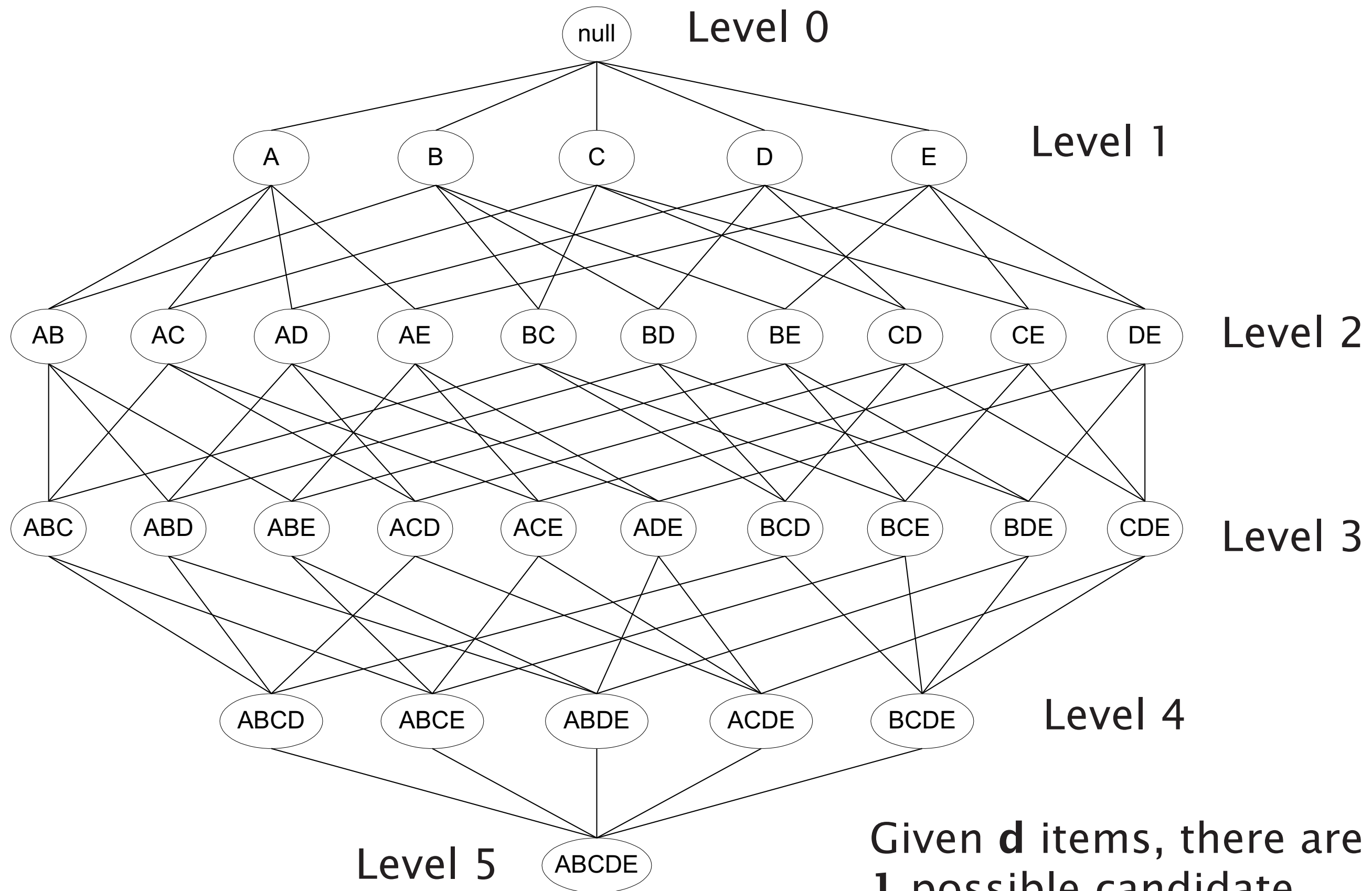
$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

- Brute-force approach:
 - List all possible association rules
 - Compute the support and confidence for each rule
 - Prune rules that fail the *min_sup* and *min_conf* thresholds

⇒ Computationally expensive!

Market Basket – Association Rule Mining

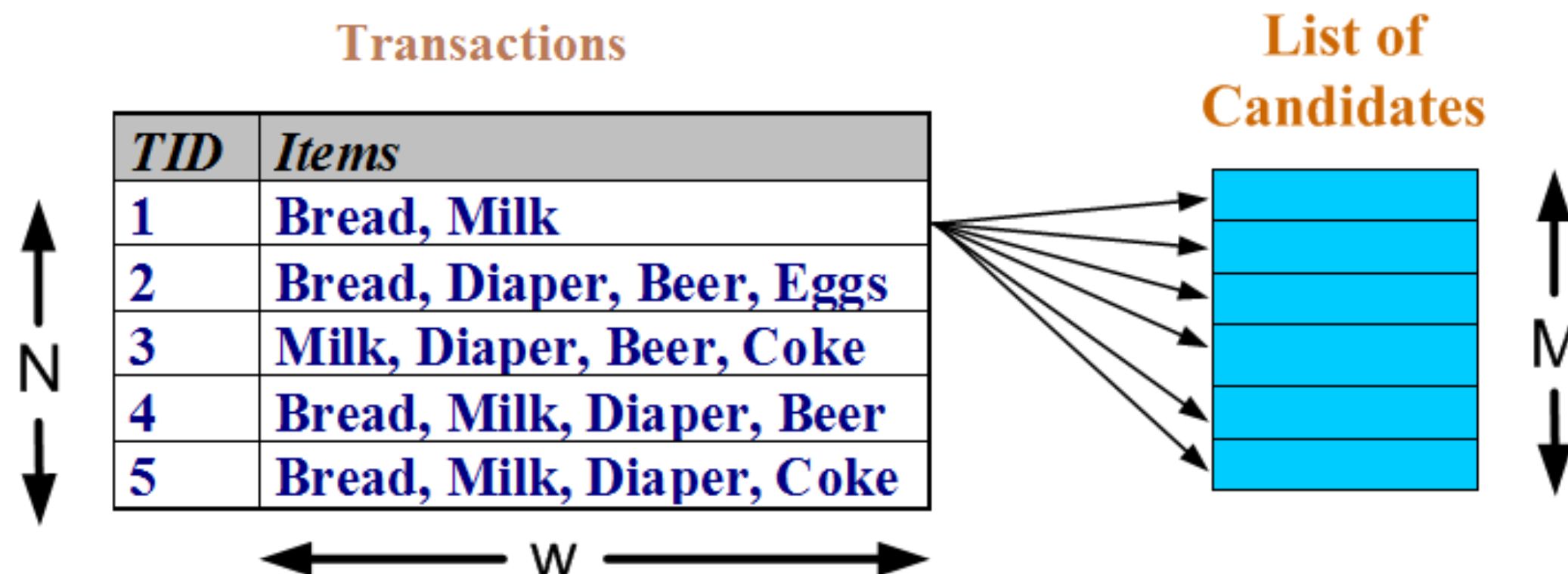


Given d items, there are $2^d - 1$ possible candidate itemsets

Market Basket – Association Rule Mining

Frequent Itemset Generation

- Brute-force approach:
 - Each itemset in the lattice is a candidate frequent itemset
 - Count the support of each candidate by scanning the database



- Match every candidate against each transaction
- Complexity $\sim O(NMw) \Rightarrow$ **Expensive** since $M = 2^d - 1$!!!*

Market Basket – Association Rule Mining

Apriori Algorithm

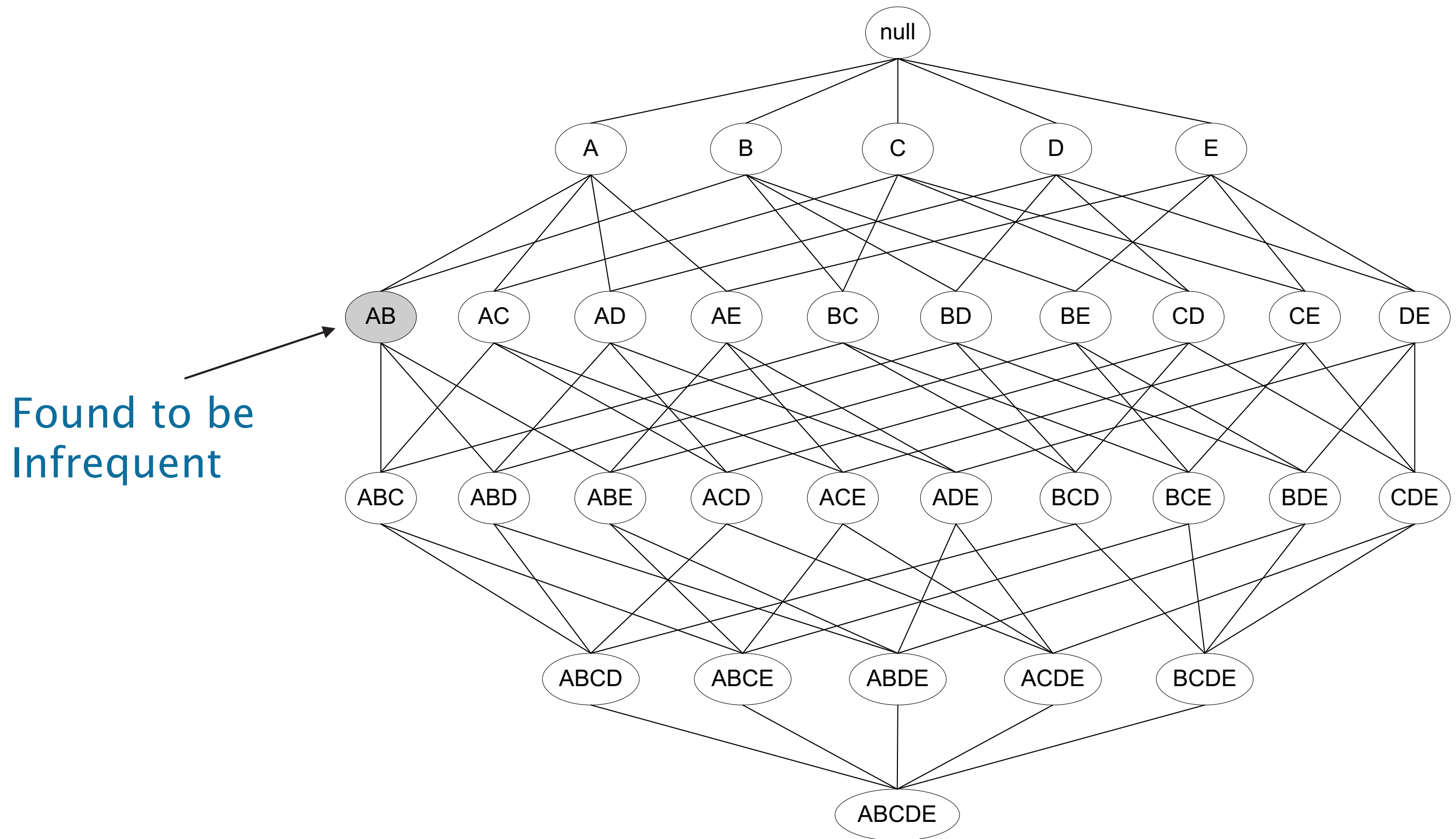
- One of the most well-known algorithms
- **Two steps** or two phases:
 - Find all itemsets that have minimum support (*frequent itemsets*, also called large itemsets)-- discover ***frequent itemsets*** from a given dataset
 - **Generate rules** from these frequent itemsets.

Market Basket – Apriori Algorithm

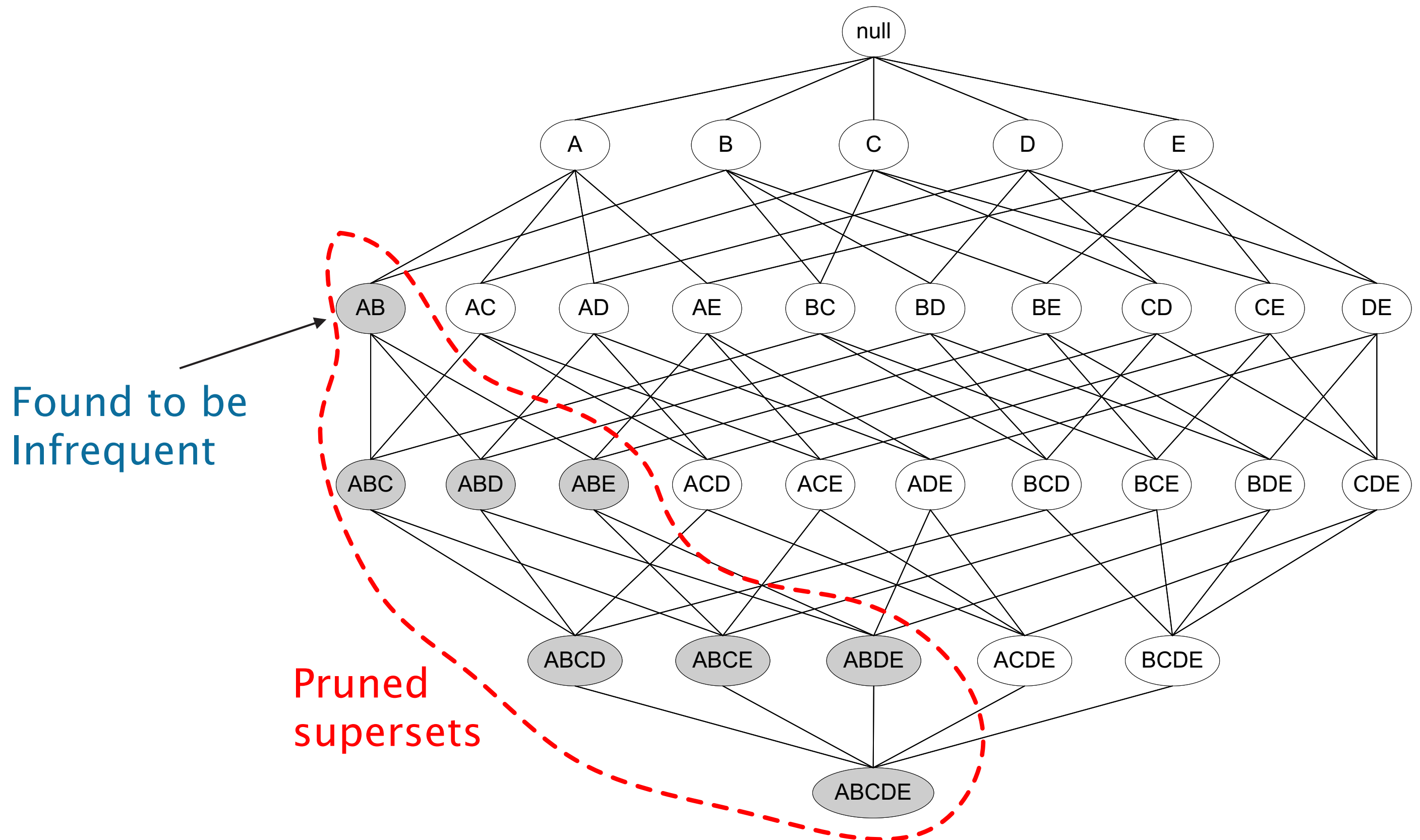
Anti-Monotone Property/ Apriori Principle

- **Any subsets of a frequent itemset must be also frequent**
 - E.g., Any transaction containing {beer, diaper, milk} also contains {beer, diaper}
 - {beer, diaper, milk} is frequent $\rightarrow$ {beer, diaper} must also be frequent
- **In other words, any superset of an infrequent itemset must also be infrequent**
 - No superset of any infrequent itemset should be generated or tested
- **Many item combinations can be pruned!**

Market Basket – Apriori Algorithm



Market Basket – Apriori Algorithm



Market Basket – Apriori Algorithm

- Method:
 - Let length of itemset be $k=1$
 - Generate frequent itemsets of length 1 (i.e., 1-itemset)
 - Repeat until no new frequent itemsets are identified
 - Generate length $(k+1)$ candidate itemsets from length k frequent itemsets
 - Count the support of each candidate by scanning the DB
 - Eliminate candidates that are infrequent, leaving only those that are frequent

Market Basket – Apriori Algorithm

Example

- Five transactions from a supermarket

TID	List of Items
1	Beer,Diaper,Baby Powder,Bread,Umbrella
2	Diaper,Baby Powder
3	Beer,Diaper,Milk
4	Diaper,Beer,Detergent
5	Beer,Milk,Coca-Cola

Market Basket – Apriori Algorithm

Example

- Min_sup 40% (2/5)

C1

Item	Support
Beer	"4/5"
Diaper	"4/5"
Baby Powder	"2/5"
Bread	"1/5"
Umbrella	"1/5"
Milk	"2/5"
Detergent	"1/5"
Coca-Cola	"1/5"

L1

Item	Support
Beer	"4/5"
Diaper	"4/5"
Baby Powder	"2/5"
Milk	"2/5"

Market Basket – Apriori Algorithm

Example

- Min_sup 40% (2/5)

C2

Item	Support
Beer, Diaper	"3/5"
Beer, Baby Powder	"1/5"
Beer, Milk	"2/5"
Diaper, Baby Powder	"2/5"
Diaper, Milk	"1/5"
Baby Powder, Milk	"0"

L2

Item	Support
Beer, Diaper	"3/5"
Beer, Milk	"2/5"
Diaper, Baby Powder	"2/5"

Market Basket – Apriori Algorithm Example

- Min_sup 40% (2/5)

C3

Item	Support
Beer, Diaper, Baby Powder	"1/5"
Beer, Diaper, Milk	"1/5"
Beer, Milk, Baby Powder	"0"
Diaper, Baby Powder, Milk	"0"

Empty

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Market Basket – Apriori Algorithm

Example

Discovery: Support $> min_sup = 40\%$

- $min_sup = 40\%$ $min_conf = 70\%$

Item	Support
Beer, Diaper	"3/5"
Beer, Milk	"2/5"
Diaper, Baby Powder	"2/5"

Generate Rules based on the searched frequent 2-itemsets

$\{Beer\} \rightarrow \{Diaper\}$, $\{Beer\} \rightarrow \{Milk\}$, $\{Diaper\} \rightarrow \{Baby Powder\}$
 $\{Diaper\} \rightarrow \{Beer\}$, $\{Milk\} \rightarrow \{Beer\}$, $\{Baby Powder\} \rightarrow \{Diaper\}$

Item	Support(A,B)	Support A	Confidence
Beer, Diaper	60%	80%	75%
Beer, Milk	40%	80%	50%
Diaper, Baby Powder	40%	80%	50%
Diaper, Beer	60%	80%	75%
Milk, Beer	40%	40%	100%
Baby Powder, Diaper	40%	40%	100%

Market Basket – Apriori Algorithm

Example

Results: Association Rules

Beer $\Rightarrow$ Diaper

- support 60%, confidence 75%

Diaper $\Rightarrow$ Beer

- support 60%, confidence 75%

Milk $\Rightarrow$ Beer

- support 40%, confidence 100%

Baby_Powder $\Rightarrow$ Diaper

- support 40%, confidence 100%

Market Basket – Summary

Terms were defined:

- ▶ **Association rules:** if X then Y , $X \implies Y$
- ▶ **Items I ,** set of all possible items i
- ▶ **Transaction:** set of items t_i such that $t_i \subset I$
- ▶ **Database D** containing all transactions $\{t_i\}_1^d$
- ▶ **Itemset:** subset of I , with k items is a k – itemset

Measures were defined:

- ▶ **Support** of itemset X is % transactions in D that contain X
- ▶ Support of Association rule $X \implies Y$ is $\frac{|t \in D; X \cup Y \subset t|}{|t \in D; X \subset t|}$
- ▶ **Confidence** is $\frac{Sup(X \cup Y)}{Sup(X)}$

Also: Final presentation does not need to be for full coursework, it is to show what you have done so far

A Priori Algorithm described